Mohammadreza Motabar

Tehran, Iran

mohammadrezamotabar@gmail.com | mrmotabar.github.io | mrmotabar | mrmotabar

Education

Master of Computer Science — *Tehran, Iran*

Sharif University of Technology — *Sep 2024 - Present*

- Current GPA: 18.43 / 20
- Advisor: Prof. S. Khazaei Co-Advisor: Prof. B. Abdolmaleki

Bachelor of Computer Science — *Tehran, Iran*

University of Tehran — *Sep 2019 - Feb 2024*

- GPA: 18.81 / 20, last two years: 19.46 / 20
- Ranked second among 66 class members in the school of computer science.
- Thesis Title: Primality Tests and Factorization Algorithms
- Thesis Advisor: Prof. A. Ghadermarzi

Research Interests

- Cryptography (especially Secure Multi-Party Computation)
- Theory of Computation and Complexity Theory
- Cyber Security, Protocol Design and Analysis
- Telecommunication Security
- Embedded Systems Security

Honors and Awards

2024 **Ranked 2nd,** National computer science master degree entrance exam (Awarded graduate tuition fee waiver)
- Offered direct admission to M.Sc. in Computer Science at University of Tehran without entrance exam
- "Excellent in term" student in all university semesters in my B.Sc. program

2022 **Third Prize,** International Mathematics Competition (IMC)

2022 **Bronze Medal,** National Mathematics Competition for university students

2020 **Top 10,** University Of Tehran's ICPC team selection

2019 **Top 1%,** National university entrance exam among 164,000 students (Awarded undergraduate tuition fee waiver)

2018 **Bronze Medal,** National Informatics Olympiad

Research Experience

Lattice Based Password-Hardened Encryption — *Tehran, Iran*

Sharif University of Technology — *Aug 2026 - Present*

- Advisors: Prof. B. Abdolmaleki, Prof. S. Khazaei, Dr. S. Nasirzadeh — *(Ongoing Project)*
- Investigating post-quantum password-hardened encryption (PHE) from lattice assumptions to defend systems against offline dictionary attacks; exploring protocol designs across core lifecycle phases (enrollment, validation, key rotation) with a focus on noise management, zero-knowledge verification, and algebraic optimization.

Primality Testing and Factorization Algorithms - Bachelor's Thesis — *Tehran, Iran*

University of Tehran — *Sep 2023 - May 2024*

- Advisor: Prof. Amir Ghadermarzi — *Link to the details*
- Delved into an extensive study of primality tests and their underlying algebra and number-theoretical foundations. Some notable areas I studied are the Fast Fourier Transform (FFT) algorithm for fast operation on numbers, the Tonelli–Shanks algorithm for computing square roots in rings, the Miller-Rabin primality test with the extension that is based on the extended Riemann hypothesis (ERH), Lucas and Grantham's Frobenius primality tests, Succinct Certificates, and AKS. Furthermore, I studied more about elliptic curves and the tests based on them, like ECM, the Shanks–Mestre method, the Schoof method, the Goldwasser–Kilian primality test, and the Atkin–Morain primality test (ECPP).
- My primary reference was the Prime Numbers book, an excellent book by R. Crandall and C. Pomerance.

Integer Factorization — *Lausanne, Switzerland*

EPFL (École polytechnique fédérale de Lausanne) - LASEC Lab - Summer@EPFL Program — *Summer 2023*

- Advisor: Dr. Tako Boris Fouotsa, Prof. Serge Vaudenay — *Link to the details*
- Briefly studied classic and modern primality tests and factorization methods, like the Miller-Rabin Test, Pollard p-1 Test, ECM, QS, and NFS. Then, delved more into the Number Field Sieve (NFS) and lots of needed theoretical materials.
- Analyzed the doubling formula in the Montgomery curves to design a new way to find two numbers with the same squares, which is used in modern and pioneered factorization methods.

Relevant Coursework

- Information Theory and Applications . . . . . (18.5 / 20)
- Advanced Introduction to Cryptography . . . . . (19 / 20)
- Introduction to Cryptography . . . . . (20 / 20)
- Descriptive Complexity . . . . . (17 / 20)
- Advanced Topics in Algorithms . . . . . (NA)
- Theory of Computation . . . . . (20 / 20)
- Linear Algebra . . . . . (20 / 20)
- Algebra 1 . . . . . (20 / 20)
- Elementary Number Theory . . . . . (20 / 20)
- Elementary Algebraic Geometry . . . . . (16.2 / 20)
- Mathematical Analysis 1 . . . . . (20 / 20)

Teaching Experience

Teaching Assistant *University of Tehran & Sharif University of Technology*

In the following courses, I was partly responsible for designing and grading assignments, projects, and exams, as well as conducting practice and problem-solving sessions.

- **Introduction to Cryptography** | Prof. S. Khazaei *Fall 2026*
- **Theory of Computation** | Prof. S. Khazaei *Spring 2026*
- **Introduction to Cryptography** | Prof. S. Khazaei *Fall 2025*
- **Theory of Computer Science** | Prof. A. Daneshgar *Spring 2025*
- **Theory of Computation** | Dr. M. Rafiee *Spring 2024*
- **Algebra 1** | Prof. A. Ghadermarzi *Fall 2023*
- **Elementary Number Theory** | Prof. A. Ghadermarzi *Fall 2023*
- **Advanced Programming** | Prof. M. Ganjtabesh *Spring 2023*
- **Theory of Computation** | Prof. M. Alizadeh *Spring 2023*
- **Compiler** | Dr. D. Tahmouresi *Spring 2023*
- **Data Structures & Algorithms** | Prof. M. Ganjtabesh *Fall 2022*
- **Fundamentals of Mathematics** | Prof. M. Mojtahedi *Fall 2022*
- **Graph Theory & Applications** | Prof. M. M. Noori *Fall 2022*
- **Fundamentals of Combinatorics** | Prof. M. M. Noori *Spring 2022*
- **Data Structures & Algorithms** | Prof. B. Babaali *Spring 2022*
- **Theory of Computation** | Prof. M. Mojtahedi *Spring 2022*
- **Data Structures & Algorithms** | Prof. B. Babaali *Fall 2021*

C++ Programming Language *Tehran, Iran*

Allameh Helli Middle School *Fall 2020 - Summer 2021*

- During the 2020 academic year, I taught C++ programming language, elementary algorithms, and data structures to some talented students at Allameh Helli Middle School.

Data Structures and Algorithms *Farzanegan High School*

- Partially, during the 2020 and 2021 academic years, besides solving algorithm problems, I taught advanced algorithms and data structures to some talented students at Farzanegan High School. I mainly used the Codeforces website problems and the book CLRS for my class.

Presentations

Multivariate Mutual Information Inspired by Secret-Key Agreement | *Summer 2026*

- Studied and presented this paper in detail, exploring one of the best-known and most natural generalizations of Shannon mutual information to multiple random variables via secret-key agreement.

One-Way Functions: Definitions and Complexity Implications | *Summer 2026*

- Studied various definitions of one-way functions, their properties, applications and relation between these definitions. Also, explored some related problems in complexity theory such as P vs UP and Isomorphism Conjecture.

Computation Through Different Lenses | *Fall 2025*

- Investigated foundational connections linking computation theory, mathematical logic, and graph theory, analyzing how computation can be represented as logical formulas (Cook–Levin, Trakhtenbrot's theorems) or paths in configuration graphs (Savitch's theorem), and how translating problems across these domains allows leveraging structural insights to establish core complexity results.

Secret Sharing | *Fall 2024 & Spring 2025*

- Explored the theoretical foundations of secret sharing for general access structures, linear schemes (LSSS), and share complexity bounds, along with key extensions including verifiable secret sharing (VSS) and their role in secure multi-party computation.

Primes is in P | *Fall 2025*

- Presented my B.Sc thesis on the topic of "Primality Tests and Factorization Algorithms" to the members of the Computer Science department at the University of Tehran.

Look Ahead on Pushdowns | *Fall 2024*

- Investigated lookahead capabilities in pushdown automata (PDA), analyzing how predicting future computation steps and configurations influences language recognition power, determinism, and computational complexity.

General Number Field Sieve | *Summer 2023* *Link to the details*

- Presented my research on Number Field Sieve to the members of the LASEC lab at EPFL University.

Projects

Classroom Object Oriented Language Compiler | Compiler Course's Project | *Spring 2022* *Link to the project*

- Implemented a compiler in C++ for the Classroom Object Oriented Language (COOL), including a syntax analyzer using Flex, a parser using Bison, and a semantic analyzer using logic inference rules.

Assembler and Disassembler | Assembly Course's Project | *Spring 2022* *Link to the project*

- Implemented an assembler and a disassembler for NASM x86 (converts Assembly commands to hexadecimal equivalents and vice versa using Python and Assembly).

Leukemia Cells Image Segmentation | Artificial Intelligence Course's Project | *Fall 2021* *Link to the project*

- Used histogram analysis, Particle Swarm Optimization (PSO), and Hill Climbing algorithm to create an image clustering algorithm and tested this algorithm on the "ALL IDB" dataset for acute lymphoblastic leukemia detection.

Basic Programmable Computer | Computer Systems Course's Project | *Spring 2021* *Link to the project*

- Implemented the 16-bit programmable computer based on Morris Mano's basic computer in Logisim.

Work Experience

Sharif Hamrah Pajoohan *Tehran, Iran*

Researcher & Developer *Sep 2024 - Present*

- Exploring enhancements and attacks on cryptographic schemes and protocols.
- Adding an encryption layer and enhanced communication protocol on a messenger app.
- Exploring and implementing attacks on mobile telecommunication network in order to find vulnerabilities and propose solutions.
- Researching on the KASUMI block cipher and Hitag2 stream cipher, its modern attacks and fastest implementation on graphic cards and FPGAs.
- Designing and Implementing a key agreement and secure communication protocol for space and ground segment of a satellite network.
- Researching on Radio Frequency Fingerprinting Identification (RFFI) with the help of deep learning.

Allameh Helli Publications *Tehran, Iran*

Book Editor *2022 - 2023*

- Assisted in authoring a sequence of instructional programming books for C++ and Python.

GreenOly *Tehran, Iran*

Full-stack Developer *Sep 2022 - Jan 2023*

- Using Laravel, Tailwind, Alpine.js, and Livewire, I developed parts of a website for educational reasons.

Certifications

- Supervised Machine Learning *Coursera / Jan 2024*
- Cryptography I *Coursera / Dec 2023*
- Introduction to TCP/IP *Coursera / Mar 2023*
- Advanced Styling with Responsive Design *Coursera / Nov 2022*
- Introduction to CSS3 *Coursera / Oct 2022*
- Interactivity with JavaScript *Coursera / Oct 2022*
- Introduction to HTML5 *Coursera / Sep 2022*
- Learning Python *Coursera / Aug 2022*
- Task-Oriented Course In Version Control With Git *Quera / Nov 2021*

Volunteer Activities

University of Tehran *Tehran, Iran*

Organizing a Competition *Winter and Spring 2024*

- Designed and conducted a series of competitions with some of my friends to increase cooperation and engagement between different majors in my school of Mathematics, Statistics, and Computer Science.

Languages

English TOEFL (91/ 120): Reading 25/30, Listening 29/30, Speaking 21/30, Writing 16/30

Persian Native